Problem 2 Written Answers: Razorpay Build-a-thon

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How to use this draft

These are first drafts of your 11 answers, written from the decisions you made while building the solution. Rewrite each one in your own voice before you submit: the brief says any claim you cannot explain counts against you. Each answer is kept between 100 and 150 words. Every number comes from P2_Churn_Scoring.xlsx at its current settings, and the last section says where each one lives.

Q1. Which problem did you pick, and who is the main user?

I picked Problem 2: merchant churn early warning and proactive outreach. The main user is the customer success manager (CSM) team, especially the lead who decides which 15 merchants get a call each month. Today CSMs often learn that a merchant is leaving only after its volume has moved to a competitor. My solution gives them a ranked list of all 100 merchants: who is at risk, the most likely reason in plain words, which playbook to run, and a drafted message to review. Two other groups rely on it. Account managers own the pricing conversations, because rate cards are their decision. Merchants receive the messages, so every message has to be accurate, polite and free of promises. Small merchants outside the 15 still get a light-touch email that a CSM approves.

Q2. What does a good outcome look like in numbers, and what does your solution change?

A good month means the 15 CSM calls reach the merchants with the most revenue at risk, before their volume moves. The data does not show how CSMs choose today, so I measure against the money at stake. Using the brief's assumptions, my 24 flagged merchants put Rs 89.7 lakh of yearly net revenue at risk. The 15 CSM calls cover Rs 83.0 lakh of it (92%) and cost Rs 22,500 of CSM time. Saving just 0.27% of that value pays for every call, so one saved Mid-size merchant covers the month's calls at least eight times over. What changes: the list is ranked every month, every flagged merchant gets some contact, and pricing and

service problems are flagged before volume falls. If a quarter of contacted merchants stayed, about Rs 20.7 lakh would be protected, but that rate is my assumption, not a result.

Q3. What does AI do, and what is deliberately left to rules or humans? Why?

Rules do the scoring. Excel formulas set the TPV alarm, the four driver flags, value at stake, the ranking and the light-touch templates. I chose rules there because the CSM team must see why a merchant was flagged, and a formula gives the same answer every time. AI does the language work. Claude tagged the 22 distinct account manager notes by driver, which I then reviewed, and drafted the 15 CSM emails from each merchant's own data. An automated check confirmed that every number in each draft appears in that merchant's row, and that no draft promises a discount, waiver, refund or date. Humans keep the decisions that carry risk. I set the thresholds and weights, a CSM approves every message before it is sent, and only an account manager can change pricing.

Q4. Which tools and models did you use, and why those?

I used Claude (Opus 5.5, in the Claude desktop app) as a research and build partner, and Microsoft Excel for the solution itself. Claude found reading material on churn, tested each of my ideas against the 100 merchants in Python, built the workbook to my rules, and drafted the outreach emails. I made the design decisions: the alarm line, which signals count for each driver, the tie-break order, ranking by value, the urgent-renewal rule and the light-touch track. I chose Excel because a CSM team already knows it, every rule is a visible formula, and settings change without code. I can move the alarm line or the cap and watch the list update. Python was used only to check the workbook: a separate Python version of my rules gave the same rank for all 100 merchants, and it caught one formula bug.

Q5. How did you test it? Show results on at least 20 cases, including the ones it got wrong.

Before writing any rules, I hand-labelled 20 merchants, drawn at random from four groups, with my churn call, driver and playbook. The finished system matches my churn call on all 20, and my driver for all 6 churners. That agreement is partly circular, because my rules grew out of those labels, so the hidden labels are the real test. Three weaknesses showed

up. First, 8 of the 20 playbooks differ: my PB-5 rule sends small, stable merchants such as C2002 to an expansion check-in, which I had labelled monitor-only, and it skips fast growers such as C2016. Second, 2 of my 6 churners, C2008 and C2015, fall below the cap and get only a light-touch email, because each is worth under Rs 1 lakh a year. Third, my alarm line was tuned on this same data.

Q6. Describe one real failure or edge case, what it could have cost, and how your design handles it.

My first version let every flagged merchant renewing within 60 days skip the queue. Testing showed that 8 such merchants took 8 of the 15 slots and pushed out C2061. C2061 had the biggest success-rate drop in the data, from 94.9% to 81.5%, and had already asked us for a review without getting a clear answer. Missing it would have risked about Rs 2 lakh a year of net revenue, from a merchant who had told us exactly what was wrong. I changed the rule so that only merchants renewing within 30 days, before next month's re-check, skip the queue. Those renewing in 31 to 60 days are caught at the next review by a separate escalation rule. C2061 is back on the CSM list, and 4 of my 6 hand-labelled churners now reach a CSM instead of 3.

Q7. What are the risks (wrong answer, bias, compliance, privacy), and what controls did you add?

Wrong answers: a false alarm costs Rs 1,500 of CSM time, so the worst case for the month is Rs 22,500. A missed churner costs far more, so every flagged merchant gets at least a light-touch email, and the list is re-checked monthly. Bias: ranking by value favours large merchants, so small ones rarely get a CSM call. I made that choice openly and gave small merchants their own light-touch track instead of ignoring them. Compliance: no message promises a discount, waiver, refund or date, an automated check looks for those words, and pricing changes stay with the account manager. WhatsApp is used only where a merchant has opted in. Privacy: the data is synthetic, and no message quotes internal signals such as logins or a risk score back to the merchant. Above all, nothing is sent until a CSM approves it.

Q8. If you had 3 months and a small team, what would you build next, and how would you know it worked?

First, I would record outcomes: which merchants were contacted, which churned and which stayed. That would replace my placeholder weights with measured save rates for each driver. Second, I would hold back a small random group of flagged merchants from light-touch outreach. The uplift reading warns that contact can push some merchants away, and a held-back group is the only way to see whether outreach really helps. Third, I would refresh the data weekly and track each merchant's signals over time, so a CSM sees risk climbing instead of a monthly snapshot. Fourth, an AI would tag new account manager notes as they are written, with a person checking each tag. I would know it worked if we catch churners weeks before their volume falls, if contacted merchants keep more volume than the held-back group, and if the top 15 hold more real churners each month.

P2.1 Which signals mattered most, and why?

Four signals did most of the work. The latest month's TPV against the merchant's own 6-month average is the symptom every churner shares. A success-rate drop was rare but decisive: 7 merchants fell by 5 points or more, and all 7 also lost volume. Unresolved tickets carried the service story. All 10 merchants with 2 or more open tickets also gave a CSAT of 2 or below, so I merged the two signals. Account manager notes were the only place pricing risk showed up. Six merchants were comparing providers or asking for rate cards, all renewing within 60 days, while their volume had fallen only 9 to 13%. That is why the pricing flag does not wait for TPV to fall. Renewal mattered as urgency, not as risk on its own. Tenure did not matter: merchants losing volume and the rest had median tenures of 34 and 34.5 months.

P2.2 How did you avoid flagging seasonal and low-volume merchants?

I compare each merchant with its own history, not with other merchants. The TPV alarm looks at the latest month against that merchant's 6-month average. Seasonal merchants dip hard, 23 to 26% below average at their lowest, but bounce back in month 6 to only 6 to 10% below. Real decliners keep falling, to 12 to 30% below, so an alarm line at -10.5% separates the two. A flag also needs a cause: a dip with no driver behind it is only watched. Low-volume merchants are safe because a percentage change ignores size, and

low logins with one product count only when TPV is also falling. Without that rule, 8 tiny but stable merchants would have been flagged. Busy-but-healthy merchants are safe because I count unresolved tickets, not total tickets. As a result, none of the 8 merchants with a seasonal note is flagged.

P2.3 How should the playbook differ for a large merchant versus a small one?

The driver decides which playbook runs; size decides how it is delivered. Value at stake ranges from Rs 0.2 lakh to Rs 27 lakh a year, so CSM time follows value. A large merchant gets a personal call and more senior people. For Dhruva Enterprises (C2004), an Enterprise merchant comparing quotes 39 days before renewal, the draft brings in the account manager and a senior leader. For Everest Learning (C2088), the CSM sets up a joint call with Payments Ops. A small merchant usually gets a light-touch email, or WhatsApp if it has opted in, filled from its own data and approved by a CSM. It moves up to a call if it replies, its signals worsen, or renewal comes within 60 days. Small merchants renewing within 30 days get a call straight away, which is why four small pricing merchants are on this month's CSM list.

Where the numbers come from

Every figure comes from P2_Churn_Scoring.xlsx at its current settings: alarm line -10.5%, urgent window 30 days, placeholder weights. If you change a setting, re-check the answers that quote it.

Figure	Used in	Where to find it
24 flagged, 15 CSM calls, 9 light-touch	Q2, Q7	Summary tab, Merchants
Rs 89.7 lakh at stake; Rs 83.0 lakh (92%) on the CSM list	Q2	Summary tab, Value at stake
Rs 22,500 cost; 0.27% break-even; Rs 20.7 lakh at a 25% save rate	Q2, Q7	Summary tab, Cost vs value

Figure	Used in	Where to find it
20 of 20 calls, 6 of 6 drivers, 12 of 20 playbooks, 4 of 6 churners on the CSM list	Q5, Q6	Test vs labels tab, top rows
C2061: success rate 94.9% to 81.5%, about Rs 2 lakh a year	Q6	Scoring tab, row C2061
Seasonal merchants 23 to 26% below average at their lowest, 6 to 10% in month 6	P2.2	Scoring tab, filter Note tag = Seasonal
7 success-rate drops, 10 merchants with 2+ unresolved tickets, 6 pricing notes	P2.1	Scoring tab: Performance flag, Unresolved and Note tag columns
Median tenure 34 vs 34.5 months	P2.1	Checked in Python during testing; not a workbook cell
Value at stake from Rs 0.2 lakh to Rs 27 lakh	P2.3	Ranked list tab, Value at stake column

Before you submit:

- [ ] Review the 22 tags on the Note tags tab, because Q3 says you did
- [ ] Read and edit the 15 drafts on the Messages tab
- [ ] Add any other tool you used to Q4
- [ ] Rewrite each answer in your own voice, keeping it between 100 and 150 words